

Testing Materials — Answer Key

Materials and Their Properties • Level 4

A. Test results table.

Paper: not waterproof, not strong, flexible. Plastic: waterproof, strong, flexible. Metal: waterproof, strong, not flexible. Fabric: not waterproof, not strong, flexible.

Answer: Reference data used to answer Sections B and C.

B. Answer the questions using the table.

1. Which material is waterproof and strong?

Answer: Plastic and metal.

2. Which materials are flexible?

Answer: Paper, plastic and fabric are all flexible — only metal is not (the worksheet's own table shows three flexible materials, not two).

3. Which material would be best for a lunch box?

Answer: Plastic — waterproof, strong and flexible.

4. Which material would be a poor choice for a raincoat?

Answer: Paper — it is not waterproof and not strong, so it would fall apart in the rain.

C. True or False?

1. A fair test changes one thing at a time.

Answer: True

2. A waterproof material lets water through easily.

Answer: False — a waterproof material keeps water out.

3. Scientists test materials to find the best one for a job.

Answer: True

4. Paper is a good material for a raincoat.

Answer: False — paper is not waterproof.

D. Why do scientists test materials before they make things?

Answer: So they can find out a material's properties (waterproof, strong, flexible) and choose the best one for the job, making sure what they build works well and is safe to use.